

Hsin-Yuan (Samuel) Lee

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Research Interests: Multimodal learning; neurocomputational vision; LLM-based scientific reasoning

Education

University of California, San Diego Expected June 2026

B.S. in Data Science & B.S. in Cognitive Science (ML & Neural Computation)

GPA: 3.911 / 4.00 Provost's Honors (8x)

Selected Coursework: Deep Learning, Representation Learning, Probabilistic Modeling, Machine Learning, Data Mining, Systems for Scalable Analytics, NLP, Data Visualization.

Publications (Under Review)

HypoEvolve: When Genetic Algorithms Meet Multi-Agent LLMs for Scientific Hypothesis Discovery
Under review at International Conference on Machine Learning (ICML) 2026.

A Mechanistic Explanation for the Inverted Face Effect
Under review at Cognitive Science (CogSci) 2026.

Research Experience

Gary's Unbelievable Research Unit (Prof. Garrison Cottrell), UCSD Aug 2025 – Present
Undergraduate Research Assistant

- Developing a neurocomputational vision model using fixation-based sampling and log-polar transforms to improve the biological plausibility of spatial encoding.
- Implementing a salience-weighted preprocessing pipeline capturing configural vs. featural structure in face perception.
- Demonstrating the Face Inversion Effect in model behavior, aligning outputs with expertise-dependent human visual processing.
- Building reproducible evaluation scripts to measure inversion robustness under expertise variations.

Qualcomm Institute (Calit2) (Dr. Neil Smith), UCSD Aug 2025 – Present
Research and Development Intern

- Built interactive LLM-driven 3D avatars integrating language, speech, and gesture generation in UE 5.
- Designed a modular LLM-TTS-Animation pipeline (FastAPI + Kubernetes) enabling real-time multimodal interaction.
- Reduced avatar response latency by ~25% through pipeline-level optimizations and cloud deployment.
- Conducting gesture synchrony and prosody alignment experiments supporting early-stage embodied AI research.

Industry Experience

Advantech Co., Ltd., Taipei, Taiwan Jun 2023 – Sep 2023

Software Development Intern

- Built a production-grade RAG chatbot using Azure OpenAI, reducing customer support response time from days to <20 seconds.
- Designed a punctuation-aware Mandarin tokenizer, improving retrieval accuracy for ambiguous queries.
- Automated ingestion and update pipelines ensuring continuous refresh of the search index.
- Collaborated with frontend team to deploy a polished UI; selected for **Best Technical Intern Project**.

Projects

Weak-to-Strong Generalization

Sep 2025 – Dec 2025

- Designed controlled experiments to measure Performance Gain Recovered (PGR) in cross-model supervision settings.
- Benchmarked GPT-4.x/5 models under noisy or weak supervision to analyze successful generalization.
- Implemented an evaluation pipeline (async API calls, JSON logs, reproducible metrics) for model-to-model supervision studies.
- Found that structure-preserving feedback improves performance over direct answer-level guidance.

Deep Climate Learning

Mar 2025 – Jun 2025

- Built CNN-MLP and ClimaX climate forecasting models for temperature and precipitation.
- Designed a custom area-weighted loss improving regional error, especially in high-variance latitudes.
- Achieved **Top 2/83 teams** with a **46% reduction** from baseline RMSE on the final leaderboard.

Teaching & Leadership

ACM @ UCSD

Director of AI Competitions

May 2025 – Present

Board Member

May 2024 – May 2024

- Led the design and organization of two campus-wide AI competitions (StarChess.ai and Blockography.AI), attracting 150+ total participants from diverse academic backgrounds.
- Introduced beginner-friendly tracks and starter notebooks, increasing novice participation by 40%.

Instructional Apprentice (TA), UCSD

Jan 2024 – Jun 2024

COGS 9: Intro to Data Science; COGS 14B: Intro to Statistical Analysis

- Led weekly sections and held office hours to support students in coding and statistical analysis.
- Graded assignments and projects, providing feedback on statistical methods and data visualization.
- Assisted instructors with course logistics and supported students during exam review sessions.

Skills

Languages: Python, SQL, JavaScript, Java, R, C++

ML/AI: PyTorch, TensorFlow, Keras, NumPy, Pandas, scikit-learn, HuggingFace, CNNs, Transformers

Tools: Git, Docker, Kubernetes, FastAPI, Flask, Jupyter, Azure OpenAI, LangChain, Unreal Engine 5

Concepts: Deep Learning, NLP, Multimodal Learning, Vision Models, Data Pipelines, Systems Design